

Overview

ShopVerse starts as a structured monolith with clear bounded-context packages. Event-driven inter-context communication via Kafka. Resilience4j guards all remote calls. gRPC for internal, GraphQL for flexible queries.

Bounded Context Boundaries

com.shopverse.catalog – Products, Categories, Search

com.shopverse.ordering – Orders, Cart, Checkout Facade

com.shopverse.inventory – Stock, Reservations

com.shopverse.notification – Email, SMS, Push

com.shopverse.search – Elasticsearch sync + query

Rule: NO direct cross-context service calls – only Kafka eve

Saga Pattern (PlaceOrderSaga – Choreography)

OrderPlaced (ordering) ■■→ Kafka: orders.placed

StockReserved / StockInsufficient (inventory) ■■→ Kafka

PaymentCaptured / PaymentFailed (payment) ■■→ Kafka

OrderConfirmed ✓ OR Compensating events ↓

PaymentFailed ■■→ StockReleased ■■→ OrderCancelled

Resilience4j Configuration

Pattern	Annotation	Class	Config
Circuit Breaker	@CircuitBreaker(name="payment")	PaymentService.charge()	failureRateThreshold=50%
Retry	@Retry(name="inventory")	InventoryService.reserve()	Default 3 attempts
Rate Limiter	@RateLimiter(name="search")	ProductSearchService.search()	Configurable in yml
Bulkhead	@Bulkhead(name="notifications")	NotificationService.send()	Thread pool isolation

API Technology Comparison

Technology	Endpoint	Use Case	Format
REST (Spring MVC)	GET/POST /api/v1/**	Primary external API	JSON / XML

Technology	Endpoint	Use Case	Format
gRPC	CheckStock(request)	Internal inventory check (order ingestion unit)	Protobuf (binary)
GraphQL	GET /graphql	Flexible frontend queries	JSON over HTTP
SSE (WebFlux)	GET /reactive/products	Streaming product catalog	text/event-stream

Testing Strategies

Level	Tool / Annotation	Class
Unit	@ExtendWith(MockitoExtension)	Every service
Component	@SpringBootTest + WireMock	All external deps mocked
Contract (Consumer)	Pact – OrderingConsumerTest	Defines inventory-service contract
Contract (Provider)	Pact – InventoryProviderVerificationTest	Verifies against contract
Integration	Testcontainers AbstractIntegrationTest	Real DB, Redis, Kafka containers